

And if you know it, you can start with Arduino. You do not need any knowledge of AtmegaXX8 controllers' architecture and can start with only the basic knowledge of C, C++ and JAVA. The Arduino integrated development environment (IDE) is as easy as writing a program in these basic languages. And that's why it is popular amongst software developers and hobbyists too."

2 An edge over other prototyping platforms

There are many factors that make Arduino popular compared to its proprietary and Open Source counterparts.

Wiring—another Open Source platform—is older than Arduino (launched in 2003) and simple too but it did not become popular like Arduino. One of the key reasons is that many Arduino boards (list available on <http://arduino.cc/en/Main/Boards>) are available with support from the Arduino community. So designers can select the boards as per their needs, whereas choice is less in Wiring.

"Currently, Arduino and BeagleBoard are the two most popular boards in India, though none of these is designed in India," opines Chatterjee.

BeagleBoard, which is a single-board computer, is fast and powerful as it comes packed with a DM3730CBP 1GHz processor (commonly used in most smartphones), five USB 2.0 ports, 512 MB of memory and on-board ethernet.

"Compared to Arduino, which is available as an 8-bit, 16-bit or 32-bit microcontroller board, BeagleBoard is more advanced. But in a way, this has become its drawback also" says Jadhao. He explains, "It is a bit complex to get started with BeagleBoard. If you need to design any advanced application, Arduino is not the answer. In such situations, working on BeagleBoard will help, as this single-board computer is designed for performing most of the needed applications without adding extra hardware."

Links for Arduino do-it-yourself projects

- > <http://tenettech.com/comet/?p=386>
- > <http://hacknmod.com/hack/top-40-arduino-projects-of-the-web/>
- > <http://www.instructables.com/technology/arduino/>
- > <http://hackaday.com/category/arduino-hacks/>
- > <http://luckylarry.co.uk/>

However, with the availability of powerful Arduino solutions, such as the PIC 32-bit Arduino solution of Microchip named chipKit, which according to the company is the first-ever Arduino-compatible microcontroller-based Open Source development platform, even this requirement has been addressed to some extent. PIC32-based chipKit boards enable 80MHz performance, and provide up to 512kB Flash and 128kB RAM. The boards feature connectivity peripherals like Ethernet, controller area network and USB (full-speed host, device and OTG), plus peripherals such as multiple timers, a 16-channel I2S analogue-to-digital converter (ADC), two comparators, multiple I²C, serial peripheral interface, and universal asynchronous receiver/transmitter interface. chipKit platform also features MIPS Technologies' MIPS32 M4K core.

"Apart from this, various Arduino shields are available (for simple to complex project requirements) to integrate the required functionalities on the Arduino board via external connectors. Besides, since its microcontroller is Open Source, it is easy to clone Arduino as per different requirements, giving designers and hobbyists a lot of flexibility in determining the components to be used on the platform. It even ensures that the cost of development is within their budgets. But this is not the case with BeagleBoard, which is managed by Texas Instruments and/or its partners," says Sani Theo, an electronics engineer.

"Another flip side of BeagleBoard is that its documentation and tutorials are not well organised and BeagleBoard community is not as strong as

the Arduino community," observes Jadhao.

However, Ram, technical evangelist and writer, Tenet Technetronics, has another perspective to share in this regard: "While proprietary hardware help in maintaining a contained quality and delivery model, the amount of creativity of these tools is quite low, which unfortunately is the hard fact. A growing inclination towards making designs open is also found in the recent past, wherein companies themselves are coming up with some kind of Open Source hardware to promote the use of their silicon."

3 Powered by design engineers from around the globe

Open Source as a concept and philosophy is driven by the core principle of putting the power in the hands of the user. Since Arduino too is an open platform, it is being used by a large number of developers and users across the globe, many of whom are design engineers.

"By giving the CAD file in the hands of many design engineers around the world, you simply make your own platform more powerful without directly hiring them. That is the core reason why Arduino is so successful even as a business model," observes Chatterjee.

"The number of advanced users and developers who are using Arduino is already huge compared to many other platforms, and this number is growing," he adds.

4 A dedicated and well-focused community

"Most design engineers say that Arduino has become popular due to the